

Campus Food Ordering System

Develop a mini food-ordering platform for a college cafeteria where students can browse food items and place orders.

Objectives

The objective of this project is to develop a mini full-stack food ordering system for a college cafeteria that allows students to browse menu items, manage a cart, place orders, and track their order status. The system should also provide an admin interface for managing food items and student orders.

The system should accomplish the following:

1. User Authentication
 - Allow students to register using basic details.
 - Allow registered students to log in and log out.
 - Maintain separate access for students and administrators.
2. Food Menu Display
 - Display all available food items with:
 - Food name
 - Price
 - Category
 - Availability
 - Image, if implemented
 - Organize food items into categories such as Snacks, Beverages, Main Course, etc.
3. Cart Management
 - Allow users to add food items to the cart.
 - Allow users to increase or decrease item quantity.
 - Allow users to remove items from the cart.
 - Automatically calculate:
 - Quantity of each item
 - Total price for each item
 - Overall/grand total
4. Order Placement
 - Allow users to confirm and place an order from the cart.
 - Generate a unique Order ID for every successful order.
 - Store order details in the database.

5. Order Tracking

- Display the current status of an order.
- Follow the order flow:

Placed → Preparing → Ready → Completed

6. Order History

- Allow students to view their previously placed orders.
- Display details such as:
 - Order ID
 - Ordered items
 - Total amount
 - Order date/time
 - Current/final status

7. Admin Food Management

- Allow the administrator to:
 - Add new food items
 - Update existing food items
 - Delete food items
 - Change food availability
 - Manage food categories

8. Admin Order Management

- Display all student orders to the administrator.
- Allow the administrator to update the order status from:
 - Placed
 - Preparing
 - Ready
 - Completed

9. Database Integration

- Store user information, menu items, cart/order details, and order history in a database.
- Retrieve and update data dynamically through the application.

10. User-Friendly Interface

- Develop a simple and responsive interface.
- Provide proper navigation between menu, cart, orders, login, and admin pages.

Expected Output

At the end of the project, a working Campus Food Ordering System should be available with separate student and admin functionalities.

For the student side, the expected output should include a registration/login page, cafeteria menu with categorized food items, functional shopping cart, automatic bill calculation, order confirmation with a unique Order ID, current order status, and order history.

For the admin side, the expected output should include an admin dashboard where the administrator can add, edit, or delete food items and view incoming orders. The administrator should also be able to change an order's status from Placed → Preparing → Ready → Completed.

A successful implementation can be demonstrated using a flow such as:

Student Login → Browse Menu → Select Food → Add to Cart → Update Quantity → View Grand Total → Place Order → Generate Order ID → Track Order → View Order History

and

Admin Login → Manage Menu → View Orders → Update Order Status